Fuzzy rho-calculus: a quantale-graded interaction,
and its lift to ci GSLTs with binding

A working note

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Abstract

We present a concise axiomatization of a fuzzy rho-calculus and a minimal model: the comm rule transmits not Q but $(u \circ x) * Q$, where $u \circ x$ is a quantale element measuring the match between the two interaction surfaces and $a * (-)$ is its action on terms. We compute the model in full, show that the order axiom is a *theorem* there (modulo one congruence side-condition isolated as a lemma), and observe that the model commits to an *accumulator* reading of $\circ$, with a dual *attenuator* reading available. We then sketch the intended payoff: this is the flat-substitution, asymmetric, single-surface instance of a general construction — a quantale-graded interaction in a continued–interactive GSLT — and identify what each ingredient must become once continuations carry binding, in both the asymmetric ("usual suspects") and the symmetric (unification/fusion) cases. The slogan that emerges: the comparison $\circ$ is *imposed* in flat rho but *generated* by unification in the symmetric lift, where A is (a quotient of) the substitution lattice and $\circ$ is mgu.

1 Recap: rho and its fuzzification

Recall the reflective syntax of the rho-calculus [1], with processes P and names x mutually defined, and the single structural reduction (comm):

$$P, Q \ ::= \ \mathbf{0} \ | \ \mathsf{for}(y \leftarrow x)P \ | \ x!(Q) \ | \ P \mid Q \ | \ *x, \qquad x, y \ ::= \ @P,$$

$$\mathsf{for}(y \leftarrow x)P \ | \ x!(Q) \ \longrightarrow \ P\{@Q/y\}.$$

Here $@P$ quotes a process to a name and $*x$ drops a name back to a process; we write $\overline{x}$ for the dereference, or unquote (also written $\#x$), so $\overline{@P} = P$ and $@\overline{x} = x$.

Notational conventions. We use Proc to denote the collection of all process terms generated by the grammar, and N (or on occasion $@\mathsf{Proc}$) to denote the collection of all names generated by the grammar.

The fuzzy data. Fuzzification is purely semantic. One adds

$$(- * -) \colon A \times \mathsf{Proc} \to \mathsf{Proc} \quad \text{(action of grades on processes)}, \qquad (- \circ -) \colon \mathsf{N} \times \mathsf{N} \to A \quad \text{(surface comparison)},$$

and replaces strict channel equality in comm by a graded comm in which the transmitted term is modulated by how well the two channels match:

$$\mathsf{for}(y \leftarrow x)P \ | \ u!(Q) \ \longrightarrow \ P\left\{ @\big((u \circ x) * Q\big) \,/\, y \right\}. \tag{1}$$

Read $u \circ x$ as "how closely u and x match" and $(u \circ x) * Q$ as "the part of (the information in) Q that gets through." Classical rho is the special case where $\circ$ returns a unit grade on a perfect match and $a * (-)$ is then the identity.

The grade structure. We demand that A carries a quantale structure [5, 6] $(A, \leq, \otimes, \mathbf{1})$ — a complete lattice with an associative product $\otimes$ that distributes over arbitrary joins — and a *measure*

$$\mu\colon \mathsf{Proc} \to A,$$

subject to the

Order axiom. $a_1 \leq a_2 \implies \mu(a_1 * P) \leq \mu(a_2 * P)$.

A bigger match grade must read off as at least as much transmitted information.

2 A minimal model

The minimal model takes grades to be names themselves and realises every piece through quote/par/dereference:

$$\begin{aligned} A &= \mathsf{N}, & x_1 \circ x_2 &= @\big(\,\overline{x_1} \mid \overline{x_2}\,\big), \\ \mu(P) &= @P, & a * P &= P\big\{\,(a \circ z)/z \;:\; z \in \mathrm{FN}(P)\,\big\}. \end{aligned} \tag{2}$$

The product is par-under-quote, the unit is $@\mathbf{0}$ (since $P \mid \mathbf{0} \equiv P$), the order is sub-multiset inclusion of par-components, and joins are componentwise suprema of multiplicities. Par is the *product*, not the join; that distribution of $+$ over max is exactly what makes this a quantale rather than a mere monoid.

2.1 Unfolding the action

Fix u, x and Q with $\mathrm{FN}(Q) = \{z_1, \ldots, z_k\}$. Put $a = u \circ x = @(\,\overline{u} \mid \overline{x}\,)$, so $\overline{a} = \overline{u} \mid \overline{x}$. For each z_j,

$$a \circ z_j \;=\; @\big(\,\overline{a} \mid \overline{z_j}\,\big) \;=\; @\big(\,\overline{u} \mid \overline{x} \mid \overline{z_j}\,\big).$$

Hence the action is a uniform *re-channelling*, never a deletion:

$$(u \circ x) * Q \;=\; Q\Big\{\,@\big(\,\overline{u} \mid \overline{x} \mid \overline{z}\,\big)/\,z \;:\; z \in \mathrm{FN}(Q)\Big\}. \tag{3}$$

Every channel z on which Q acts is stamped with the *match signature* $\overline{u} \mid \overline{x}$ of the redex; the whole term structure of Q survives intact.

2.2 The two redexes

With the example's orientation (input on u_i, output on x_i; commutativity of $\circ$ makes the orientation immaterial, since $@(\,\overline{u} \mid \overline{x}\,) \equiv @(\,\overline{x} \mid \overline{u}\,)$), equation (3) gives

$$\begin{aligned} \mathsf{for}(y \leftarrow u_1)P \mid x_1!(Q) \;&\longrightarrow\; P\Big\{\,@\big(Q\{@(\,\overline{u_1} \mid \overline{x_1} \mid \overline{z}\,)/z\}\big)/\,y\Big\}, \\ \mathsf{for}(y \leftarrow u_2)P \mid x_2!(Q) \;&\longrightarrow\; P\Big\{\,@\big(Q\{@(\,\overline{u_2} \mid \overline{x_2} \mid \overline{z}\,)/z\}\big)/\,y\Big\}. \end{aligned}$$

The two residuals differ in exactly one place: the signature $\overline{u_1} \mid \overline{x_1}$ versus $\overline{u_2} \mid \overline{x_2}$ stamped into every channel of Q.

2.3 The order axiom is a theorem here

Lemma 1 (Congruence side-condition)**.** *Let the order $\leq$ on $A = \mathsf{N}$ be the least pre-congruence for the rho operators with $@\mathbf{0}$ as bottom (equivalently: sub-multiset inclusion at the top level, propagated through prefixes $\mathsf{for}(\cdot)$, par, output, and quote). Then name-substitution into a fixed context is monotone: $a \leq a'$ implies $C[a] \leq C[a']$ for every term context C.*

The point of Lemma 1 is that a prefix $\mathsf{for}(y \leftarrow x)P$ is a single molecule; sub-multiset comparison at the top level alone does *not* push through it, and without congruent propagation the monotonicity below leaks precisely at the binders.

Proposition 1 (Grade monotonicity)**.** *In the model* (2) *with the order of Lemma 1, the order axiom holds. Consequently, if $\overline{u_1} \mid \overline{x_1} \sqsubseteq \overline{u_2} \mid \overline{x_2}$ then $\mu\big((u_1 \circ x_1) * Q\big) \leq \mu\big((u_2 \circ x_2) * Q\big)$, i.e. the redex with the larger match signature carries at least as much information into the residual.*

Proof. $\circ$ is $+$ on multisets, hence monotone in each argument; by Lemma 1 substitution of a $\leq$-larger grade into the fixed context $Q\{(-)/z\}$ yields a $\leq$-larger term, so $(u_1 \circ x_1) * Q \sqsubseteq (u_2 \circ x_2) * Q$. Applying $\mu = @(-)$, which is monotone by construction, gives the claim. The order axiom is the same statement with $u \circ x$ abstracted to a free grade a. □

So in this model monotonicity is not an extra demand but a consequence of $\circ$ being the (monotone) quantale product and μ a monotone reading of it.

2.4 What kind of "fuzzy" is this? Accumulator vs. attenuator

Remark 1 (The model commits to an accumulator). With $\circ = @(\,\overline{u} \mid \overline{x}\,)$ the construction is an *accumulator*, not an attenuator:

- it never deletes any of Q — by (3) the action only renames channels, so all term structure of Q always survives;
- $u \circ x$ grows when *either* side is large, regardless of overlap, so it measures combined bulk, not similarity;
- even a perfect match $u = x$ gives $u \circ x = @(\,\overline{x} \mid \overline{x}\,)$, a doubling — a *watermark*, not the identity. Classical rho is recovered only at $u = x = @\mathbf{0}$.

If one wants the literal "distance/attenuation" reading — perfect match transmits cleanly, mismatch transmits less — the natural swap is the multiset *meet* $u \circ x = @(\,\overline{u} \wedge \overline{x}\,)$, the shared sub-process. Then $u \circ x$ grows with genuine overlap, and because meet can map distinct channels $@Z_j$ to a common stamped name, channels can *collapse*: attenuation as lost distinguishability rather than lost terms. Meet over multisets is still a (frame-like) quantale and $\mu = @(-)$ stays monotone, so nothing breaks.

Remark 2 (A foretaste of the lift: mgu vs. lgg). The fork in Remark 1 is not ad hoc. In the subsumption order on terms, the accumulator ("keep everything from both surfaces") is aligned with *unification* [7] — the most general common *instance*, mgu — while the attenuator ("keep only what is shared") is aligned with *anti-unification* [8, 9] — the least general common *generalization*, lgg. These are dual operations in the same lattice. Section 3 takes this seriously: when continuations carry binding and interaction is symmetric, the comparison $\circ$ *is* a (anti-)unifier and need not be invented.

3 Lifting to ci GSLTs with binding

Rho is not binding-free — the input prefix $\mathsf{for}(y \leftarrow x)P$ binds y in P — but it has only that one binding form, and in the model of §2 it is dispatched by ordinary capture-avoiding substitution: the action (2) acts on $\mathrm{FN}(P)$ as a flat set of free names, never needing to be transported across y in any non-trivial way. (Rho also lacks a ν-style scoping operator, so there is no scope to extrude.) In a continued–interactive GSLT, by contrast, a *language* is presented by a grammar (sorts/operations, several of which may bind), equations (structural congruence), and rewrites (reduction) [2, 3, 4], with arbitrary declared binders, and the continuation in an interaction redex *carries binding structure.* The whole graded apparatus must be reread inside the category of terms-with-binding — nominal sets / a presheaf category over name-contexts, or de Bruijn families [14, 15, 16]. We list what each of the three ingredients becomes; the flat rho action is then the degenerate case in which the transport law below is trivial.

3.1 (i) The action becomes a graded action respecting binders

$a * (-)$ is no longer a flat substitution but an A-graded, capture-avoiding endofunctor on the term presheaf. The defining law is that it commutes with the binding (context-extension) functor: writing $[y]\,P$ for the abstraction of y in P (whatever the binding form — rho's input prefix, λ, a ν-restriction),

$$a * ([y]\,P) \;=\; [y]\,(a' * P),$$

where a' is a transported across the binder (in nominal terms, a taken with support away from y, α-renaming as needed). This transport law is the precise content of "continuations have binding structure." The order axiom lifts unchanged in statement: $*$ is a lax action of the quantale (functorial and monotone in A), in the sense of graded monads [17], and μ a lax monotone measure, whence *fuzzy reduction is grade-monotone* — now internal to the presheaf category.

3.2 (ii) The comparison $\circ$: subject-matching vs. unification

Asymmetric case (the usual suspects: λ [11], π [10], rho). There is a privileged binding surface. The surfaces are a subject pair (the input channel x, the output channel u); $\circ$ compares subjects, $u \circ x \in A$; and the action modulates the transmitted term before a *one-directional* graded substitution into the binder body, exactly as in (1). This is a near-verbatim lift: only the substitution becomes the presheaf's capture-avoiding one, and the action must be transported under any binders occurring inside the transmitted term.

Symmetric case (fusion/solos via unification). There is no privileged binder. Both surfaces present terms with (meta)variables, and the reaction is a *fusion* [12, 13]: it computes a unifier $\sigma = \mathrm{mgu}(\mathrm{surface}_1, \mathrm{surface}_2)$ and applies σ to *all* continuations in scope, each application a graded action. The grade is read off the unifier,

$$\circ(\mathrm{surface}_1, \mathrm{surface}_2) \;=\; \mathrm{grade}(\sigma) \in A,$$

with natural choices ranging from the crudest two-element quantale $\{\top \text{ if unifiable, } \perp \text{ otherwise}\}$ ("fuzzy" = "does it react at all") to a rich quantale built from the *substitution lattice* — substitutions ordered by instantiation, mgu as its meet/join — with $\mathrm{grade}(\sigma)$ a monotone invariant such as the number of equations discharged, the term-size of σ, or μ of the substituted material.

This is the conceptual payoff. In flat rho the comparison $\circ$ had to be *imposed* as extra structure (par-under-quote). Under symmetric interaction it is *generated*: the substitution lattice already is the requisite ordered structure, mgu already is its meet/join, so A may be taken to be (a quotient of) that lattice and $\circ$ to be mgu itself. The intuition "how far apart u and x are" then becomes literal — the *unification distance*: how much the mgu had to do to make the surfaces agree, or whether they agree at all. Dually (Remark 2), the attenuator/meet variant is realised by anti-unification: $\circ = \mathrm{lgg}$ keeps the shared skeleton of the two surfaces, the symmetric surface of the $@(\,\overline{u} \wedge \overline{x}\,)$ swap.

3.3 (iii) Side conditions: scope extrusion as a precongruence law

Fusion extrudes scope: a fused name may sit under a restriction, and σ must respect it (the classical fusion side-condition — do not fuse names trapped under conflicting binders; push the restriction outward instead). The graded version adds a stability requirement that is the symmetric analogue of Lemma 1:

$$\mathrm{grade}(\sigma) \text{ must be stable under } \alpha\text{-conversion and scope extrusion,}$$

i.e. computing the grade before and after pushing a ν outward must agree (up to the action's transport law). Equivalently, the order on A must be a pre-congruence for the binders, not merely for par; otherwise grade-monotonicity leaks exactly at the binders — the same failure Lemma 1 forecloses in the flat, top-level case.

4 Summary and open ends

Summary. Fuzzy rho is the flat-substitution, asymmetric, single-surface, par-quote instance of one construction: a quantale-graded interaction in a ci GSLT, with the grade obtained by comparing interaction surfaces (subject-matching when asymmetric, (anti-)unification when symmetric), acting capture-avoidingly on continuations, and grade-monotonicity guaranteed by a precongruent order. The two warm-up redexes of §2 are the shadow this construction casts when binding is trivial and the comparison is par.

Open ends.

1. Which monotone invariant of the mgu yields a μ for which grade-monotonicity says something about *genuinely transmitted* information, rather than mere substitution bulk?
2. Make the mgu/lgg duality of Remark 2 a precise adjunction between the accumulator and attenuator gradings, internal to the substitution lattice.
3. Reading the grade as a potential on the reduction graph (the energetics picture), is grade-monotonicity along reduction a second-law-flavoured statement, and does the precongruence condition correspond to a conservation law at the binders?

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